Project: Worked examples
Title: Insulation thickness (ISO 12241)
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1. Scope and method

Calculates the insulation thickness of a pipe or flat surface that meets a heat-loss limit, a hot-surface limit or an anti-condensation requirement. For a chosen thickness it gives the heat flow, surface temperature and PASS or FAIL.

WHAT IT CALCULATES	RELATION
Pipe resistance per metre	$R = \ln(D_e/D_i)/(2\pi\cdot\lambda) + 1/(h_{se}\cdot\pi\cdot D_e)$, $D_e = D_i + 2t$; with the inner film and pipe wall $R_{oth} = 1/(h_i\cdot\pi\cdot D_p) + \ln(D_i/D_p)/(2\pi\cdot\lambda_w)$ added in series
Flat resistance per square metre	$R = t/\lambda + 1/h_{se}$ (+ $1/h_i + t_w/\lambda_w$)
Heat flow	$q = (\theta_i - \theta_a)/R$, positive from the process to the ambient
Surface temperature	$\theta_{se} = \theta_a + q/(h_{se}\cdot\pi\cdot D_e)$ (pipe), $\theta_a + q/h_{se}$ (flat)
Surface coefficient	h_{se} as given, or $h_{se} = h_c + h_r$ with $h_r = \varepsilon\cdot\sigma\cdot(T_s^2 + T_a^2)(T_s + T_a)$; h_c from free convection to air (laminar or turbulent, the larger), forced convection (cross-flow cylinder or flat plate) and their cube-law blend; solved with θ_{se}, which it depends on
Heat-loss limit	the thickness beyond which $\lvert q\rvert \le q_{max}$; surface-temperature limit (hot): $\theta_{se} \le \theta_{se,max}$; anti-condensation (cold): $\theta_{se} \ge \theta_{dew}$ + allowance, θ_{dew} entered (for example from the psychrometrics calculation) or from the Magnus relation at θ_a and RH
Thickness solution	each criterion's thickness by scanning t from 0 to 1 m for the last broken point and bisecting the bracket after it; governing t = max over the selected criteria
Critical diameter	$D_{crit} = 2\lambda/h_{se}$; below it, insulation raises the heat flow until D_e passes D_{crit}

Basis: Calculation rules for thermal insulation of equipment and installations (clause numbers to be checked)[1]; External forced, free and mixed convection correlations[2](§7.2, §7.4, §9.6, §9.9, 7th ed. (section numbers to be checked)); Magnus saturation vapour pressure (shared with the Glaser condensation calculation)[3](cl. 6 (clause number to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
geometry	Surface	pipe (cylinder)		assumption: A round pipe.	any
h_{mode}	Where the outer surface coefficient comes from	h_se as given		assumption: A surface coefficient is given, not computed.	any
D_i	Pipe outer diameter, the inside of the insulation (pipe)	90	mm	drawing: M-PIP-014 rev A	10 … 1000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_{surf}	Height or flow length of the flat surface (correlation, flat)	1	m	assumption: Not used: a pipe.	0.1 … 10 m
t_{chosen}	Chosen insulation thickness (optional; blank = report at the required thickness, with no checks; with it, nothing blocks)	25	mm	spec: Pipework schedule M-SCH-014: 25 mm sleeves	0 … 300 mm
λ	Thermal conductivity of the insulation	0.035	W/(m·K)	assumption: Closed-cell elastomeric insulation; take the maker's value at the mean temperature (W3 material)	0.015 … 0.2 W/(m·K)
T_{proc}	Process (fluid) temperature	6	°C	spec: SPEC-M-11 §3: chilled-water flow temperature	-50 … 400 °C
T_{amb}	Ambient air temperature (also the radiation sink)	28	°C	spec: SPEC-M-11 §2: plant-room design temperature	-20 … 50 °C
RH_a	Relative humidity of the ambient air (anti-condensation)	70	%	spec: SPEC-M-11 §2: plant-room design humidity	20 … 95 %
v_{wind}	Air speed over the surface (correlation; 0 = still air)	0	m/s	assumption: Not used: h_se is given.	0 … 20 m/s
h_{se}	Outer surface coefficient (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)	8	W/(m²·K)	assumption: Still air in a plant room; replace with the project's own record.	3 … 40 W/(m²·K)
eps	Emissivity of the outer surface (for the correlation; with h_{se} as given it only judges that value against condensation on a cold surface)	0.9		assumption: Painted or plastic jacket (invented for this example); with h_se given it only judges that value against condensation.	0.05 … 1
$q_{max,m}$	Heat-loss limit per metre of pipe (optional; blank = not a criterion; the generic default of 25 W/m is to be replaced with your project limit)	12	W/m	spec: SPEC-M-11 §5: heat gain not to exceed 12 W/m	1 … 1000 W/m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dew_{margin}	Allowance of the surface above the dew point, for a cold surface (optional; blank = not a criterion)	1	K	spec: SPEC-M-11 §5: surface at least 1 K above the dew point	0 ... 5 K

3. Assumptions

A1. Heat flows in steady state, one-dimensionally, through a single homogeneous layer of constant conductivity. (Formula route of the standard.)
A2. The insulation has no thermal bridges at supports, no joints and no moisture. (Formula route of the standard.)
A3. The conductivity λ is the value entered, which should be the value at the mean insulation temperature on the sheet. (Stated.)
A4. With the correlation, the outer surface coefficient is $h_{se} = h_c + h_r$. (Incropera, §7.2, §7.4, §9.6 and §9.9 (section numbers to be checked against the owner's copy).)
A5. Radiation is $h_r = \varepsilon \cdot \sigma \cdot (T_s^2 + T_a^2)(T_s + T_a)$ to surroundings at the ambient air temperature. (Incropera, §1.2.)
A6. Free convection to air uses Holman's simplified relations at atmospheric pressure, the larger of the laminar and turbulent forms. (Holman's constants: horizontal cylinder 1.32(ΔT/D)^0.25 and 1.24ΔT^(1/3), vertical surface 1.42(ΔT/L)^0.25 and 1.31ΔT^(1/3).)
A7. Forced convection uses Churchill–Bernstein for a cylinder in cross-flow and the laminar or mixed flat-plate relations along a plate. (Incropera, §7.2 and §7.4.)
A8. Air properties are fits at the film temperature, with Pr = 0.71. (Stated.)
A9. Free and forced convection blend as $(h_{nat}^3 + h_{forced}^3)^{(1/3)}$. (Stated.)
A10. The correlation is indicative, not the standard's own. (The standard's own coefficients are not reproduced.)
A11. The dew point is entered, or is the Magnus relation at the ambient temperature and humidity. (Derived; the Glaser condensation calculation shares the function.)
A12. Below 0 °C the Magnus relation is over ice, so the humidity is then taken relative to ice. (Stated; enter the dew point for sub-zero air, because weather humidity is usually relative to water.)
A13. An allowance above the dew point is the user's choice. (Stated.)
A14. Pipes are treated as horizontal and flat surfaces as vertical for free convection. (Stated.)
A15. Air conductivity is a fit within about 3 % over 250 to 400 K. (Stated.)
A16. Air viscosity is a fit within about 4 % over 250 to 400 K. (Stated.)
A17. The required thickness is the least thickness beyond which the criterion holds for every thicker layer up to 1 m. (Stated; this matters when thin insulation on a small pipe raises the heat flow (critical diameter).)
A18. No thickness table of any standard, including BS 5422, is used or shipped. (No standard's table is reproduced.)
A19. The conductivity is the user's input, from a library material or the maker's data. (No standard's table is reproduced.)
A20. geometry (Surface) is taken as pipe (cylinder). (A round pipe.)
A21. h_{mode} (Where the outer surface coefficient comes from) is taken as h_{se} as given. (A surface coefficient is given, not computed.)
A22. L_{surf} (Height or flow length of the flat surface (correlation, flat)) is taken as 1 m. (Not used: a pipe.)
A23. λ (Thermal conductivity of the insulation) is taken as 0.035 W/(m·K). (Closed-cell elastomeric insulation; take the maker's value at the mean temperature (W3 material))
A24. v_{wind} (Air speed over the surface (correlation; 0 = still air)) is taken as 0 m/s. (Not used: h_{se} is given.)
A25. h_{se} (Outer surface coefficient (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)) is taken as 8 W/(m²·K). (Still air in a plant room; replace with the project's own record.)

A26. eps (Emissivity of the outer surface (for the correlation; with h_{se} as given it only judges that value against condensation on a cold surface)) is taken as 0.9. (Painted or plastic jacket (invented for this example); with h_{se} given it only judges that value against condensation.)

4. Working

[11] T_{dew} = inverse Magnus of $RH \cdot p_{sat}(T_{amb})$ [3]((Magnus form, as in the Glaser condensation calculation))
= T_{amb} = 28 °C, RH = 70 %
= 22.012 degC
Dew point of the ambient air

[12] $t_{req,q}$ = t such that |q| ≤ 12 W/m for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 17.814 mm
Required thickness, heat-gain limit

[13] $t_{req,dew}$ = t such that θ_{se} ≥ 23.012 °C for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 13.162 mm
Required thickness, anti-condensation allowance

[14] thickness = max over the selected criteria [1]
= $t_{req,q}$ = 17.814 mm, $t_{req,dew}$ = 13.162 mm
= 17.814 mm
Governing required thickness (heat-gain limit)

[16] $D_e = D_i + 2t$
= 90 mm + 2 × 25 mm
= 140 mm
Outer diameter of the insulation, chosen thickness

[17] $R_{ins} = \ln(D_e/D_i)/(2\pi \cdot \lambda)$ [1]
= ln(140/90)/(2π × 0.035)
= 2.0091 m·K/W
Resistance of the insulation

[18] $R_{se} = 1/(h_{se} \cdot \pi \cdot D_e)$ [1]
= 1/(8 × π × 0.14 m)
= 0.28421 m·K/W
Resistance of the outer surface

[19] $R_{tot} = R_{oth} + R_{ins} + R_{se}$ [1]
= 0 + 2.0091 + 0.28421
= 2.2933 m·K/W
Total resistance per metre

[20] $q_{per,m} = |\theta_i - \theta_a| / R$ [1]
= 22 K / 2.2933
= 9.593 W/m
Heat flow per metre of pipe (heat gain, magnitude)

[21] $T_{se} = \theta_a + q/(h_{se} \cdot \pi \cdot D_e)$ [1]
= 28 °C − 9.593 / 3.5186
= 25.274 degC
Outer surface temperature

[22] $T_m = (\theta_i + \theta_{se})/2$
= (6 + 25.274)/2
= 15.637 degC
Mean temperature of the insulation: take λ at this temperature

[23] $D_{crit} = 2\lambda/h_{se}$ at the bare surface [1]
= 2 × 0.035 / 8
= 8.75 mm
Critical diameter at the bare-surface h_{se}: below it, insulation raises the heat flow

[24] $u_q = |q| / q_{max}$
$= |q| \le 12\ W/m$
$= 0.79941$
Utilisation, heat-gain limit
[25] $u_{dew} = (\theta_a - \theta_{se})/(\theta_a - (\theta_{dew} + margin))$
$= \theta_{se} \ge 23.012\ °C$
$= 0.54661$
Utilisation, anti-condensation allowance
[26] $dT_{dep} = \theta_a - \theta_{se}$
$= 28\ °C - 25.274\ °C$
$= 2.7264\ K$
Fall of the surface below the ambient air
[27] $dT_{dep,max} = \theta_a - (\theta_{dew} + allowance)$
$= 28\ °C - 23.012\ °C$
$= 4.9878\ K$
Largest fall of the surface below the ambient air

5. Results

SYMBOL	VALUE	DESCRIPTION
thickness	17.81 mm	Governing required thickness
$q_{per,m}$	9.593 W/m	Heat flow per metre (heat gain)
T_{se}	25.27 °C	Outer surface temperature
T_{dew}	22.01 °C	Dew point of the ambient air
$t_{req,q}$	17.81 mm	Required thickness, heat-gain limit
$t_{req,dew}$	13.16 mm	Required thickness, anti-condensation allowance
T_m	15.64 °C	Mean insulation temperature
R_{tot}	2.293 m·K/W	Total resistance
D_{crit}	8.750 mm	Critical diameter (at the bare-surface h_{se})
governing	"heat-gain limit"	Governing criterion
u_q	0.80	Utilisation, heat-gain limit (governing)
u_{dew}	0.55	Utilisation, anti-condensation allowance

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Chosen thickness against the heat-gain limit	9.593 W/m	12 W/m	0.80	[1]
PASS	Chosen thickness against the anti-condensation allowance	2.726 K	4.988 K	0.55	[1]

7. Applicability

WARNING records: tabulated values not taken from a reference record are λ (Material properties); h_{se} (Outer surface coefficient); eps (Surface emissivity); $q_{max,m}$, dew_{margin} (Insulation criteria). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

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Template bs.insulation@1.2.0 Content hash b8b77970e70183e1 Engine calc-engine/1.0.0
Known-answer cases passed 14/14 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. ISO 12241:2022.
2. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
3. BS EN ISO 13788:2012.